Dossier Space Pilot — v1.4 Creation Session Log

Session ID: DS-20260811-S11
Design Lead: Claude
Operator: Ken Tombs
Date: August 11, 2026
Action: Dossier Space MCP Custody Server v1.4 rebuild and clean version separation

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Context

At the close of Session 10 (DS-20260806-S10), the MCP Custody Server v1.3 was identified as non-functional in the field due to a critical architectural gap: **missing JSON-RPC protocol layer implementation.**

The server was built with:

- ✓ Ledger writer with fail-closed semantics
- ✓ Governance ledger infrastructure
- ✓ CustodyServer class with six read-verify tools
- ✓ Correct corpus path (`...enron_mail_20150507\maildir`)
- ✗ **No JSON-RPC protocol handler** — making the tools inaccessible to Claude Code via MCP

The v1.3 code contained a simple `while True: time.sleep(1)` loop with no server actually listening for or routing MCP calls.

Defect Summary

Issue: v1.3 MCP Tool Registration Incompleteness

Component	Status	Notes
LedgerWriter	✓ Functional	Tested in v1.2 smoke test (1,147 candidates, ledger chain validated)
CustodyServer	✓ Functional	All six tools (search_messages, list_folder, read_file, get_headers, hash_document, verify_hash) present and complete
Path containment	✓ Correct	Validated: `E:\AAll XPlain\Dossier Space Pilot\ENRON Data Set Test 1\enron_mail_20150507\maildir`
JSON-RPC protocol handler	✗ **Missing**	No FastMCP server initialization, no tool registration, no MCP call routing
Stdio transport	✗ **Missing**	Server did not listen on stdio for Claude Code MCP client

Root Cause: v1.3 was built from specification fragments rather than a frozen requirements document. The JSON-RPC protocol and tool registration layers were not implemented before handoff.

Lesson (Principle): "Less haste, more speed" — infrastructure layers must be specified and frozen before implementation begins. Specification fragments lead to accumulating gaps.

Resolution: v1.4 Clean Rebuild

Rather than patch v1.3 (risk of incomplete repairs re-emerging), a **clean v1.4 rebuild** was performed with:

Architecture Repairs

1. **JSON-RPC Protocol Handler** (lines 618–732)
    - `DossierMCPServer` class wrapping custody logic
    - `@server.list_tools()` handler — registers all six tools with FastMCP 1.29.0
    - `@server.call_tool()` async handler — routes incoming JSON-RPC calls to CustodyServer methods
    - Proper `ToolResult` objects with MCP-compatible response format
2. **Tool Registration with Schemas** (lines 641–691)
    - Each tool: name, description, inputSchema (JSON Schema format)
    - Claude Code can introspect tool capabilities and validate arguments
    - Parameters fully documented (custodian, keyword, date_contains, filepath, expected_hash)
3. **Async-Aware Server Loop** (lines 734–784)

- Main entry point converted to async `main()` function
- `DossierMCPServer.run()` uses `await asyncio.Event().wait()` for persistent operation
- Proper error handling and graceful shutdown (KeyboardInterrupt)

Preserved Components (No Changes)

- **LedgerWriter** (lines 88-281) — unchanged, battle-tested from v1.2
- **CustodyServer** (lines 288-672) — all six tools intact and functional
- **Governance Ledger** — fail-closed semantics, controlled value enums, ledger write strategy
- **Path Containment** — validation logic preserved
- **Configuration** — correct corpus path confirmed from v1.2 validated version

Version Separation

- **Old:** v1.3 (broken) — `SESSION_ID = "DS-20260806-S10"` (original flawed build)
- **New:** v1.4 (rebuilt) — `SESSION_ID = "DS-20260811-S11-v1.4"` (clean separation)

Reason: Prevents automated processes from attempting to patch v1.3. Ledger entries remain unambiguous. Version number is immutable in governance record.

Validation Plan (Session 11 Execution)

Before execution sessions proceed:

1. **Dependency Check**
    - Confirm FastMCP 1.29.0 installed: `pip install mcp==1.29.0`
    - Verify Python 3.14.6 / pip 26.1.2 environment
2. **Startup Validation**
    - Run server with logging enabled
    - Confirm no import errors
    - Confirm ledger directory/file paths accessible
    - Confirm corpus path validation passes
3. **Protocol Smoke Test**
    - Claude Code initiates first tool call (e.g., `search_messages`)
    - Verify JSON-RPC call routed correctly to CustodyServer
    - Verify action ID generated and logged to governance ledger
    - Verify result returned in proper MCP format
4. **Governance Ledger Validation**
    - Confirm session_start entry emitted with v1.4 designation
    - Confirm tool_call and mcp_call entries capture parameters, results, hashes

- Confirm fail-closed fallback tested (if applicable)

Governance Notes

Document is custody, not memory — this log entry captures the rationale for v1.4 in an immutable record. Future sessions can reference this when understanding version history.

Panel Separation Maintained:

- Claude (design lead) — drafted specification and protocol layer architecture
- Claude Code (subject under investigation) — will execute workstation operations
- Ken Tombs (operator) — executes all workstation commands; confirms path, dependencies, startup

Three-Axis Defensibility:

- Authority: Ken authorizes the rebuild based on identified v1.3 defect
- Transparency: Session log documents what was broken and how it was repaired
- Audit Trail: Ledger entries will record all actions under v1.4 SESSION_ID

Status

v1.4 Code: Ready for delivery
Next Step: Workstation validation (Ken to confirm startup, ledger access, path resolution)
Publication: Rebuilt server code ready for GitHub artifacts after validation passes

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Session log prepared by: Claude (Design Lead)
Reviewed by: Ken Tombs (Operator)
Date logged: August 11, 2026